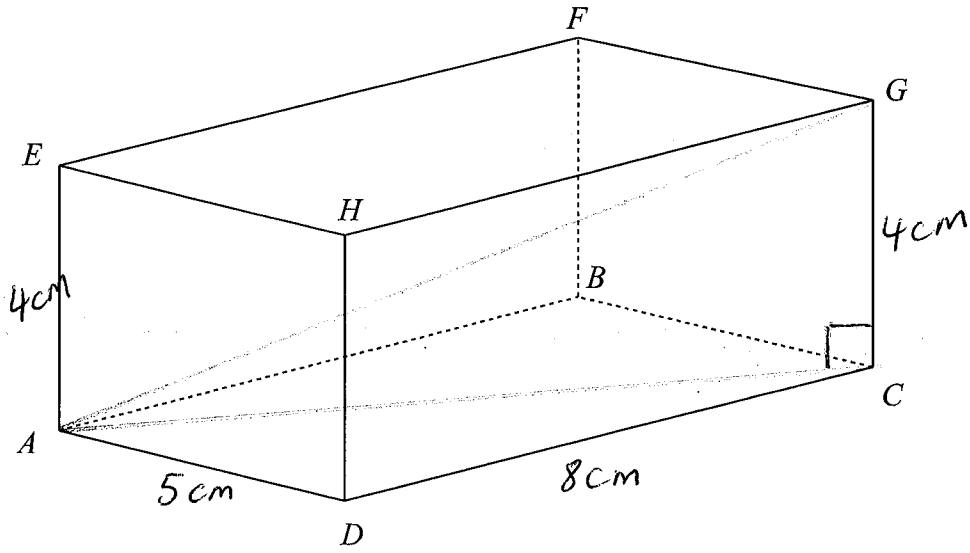
**1** The diagram shows a cuboid *ABCDEFGH*.

$AE = 4$ cm
$AD = 5$ cm
$DC = 8$ cm




Calculate the length of *AG*.
Give your answer correct to 3 significant figures.

$$AG^2 = 5^2 + 8^2 + 4^2$$

$$= \sqrt{5^2 + 8^2 + 4^2}$$

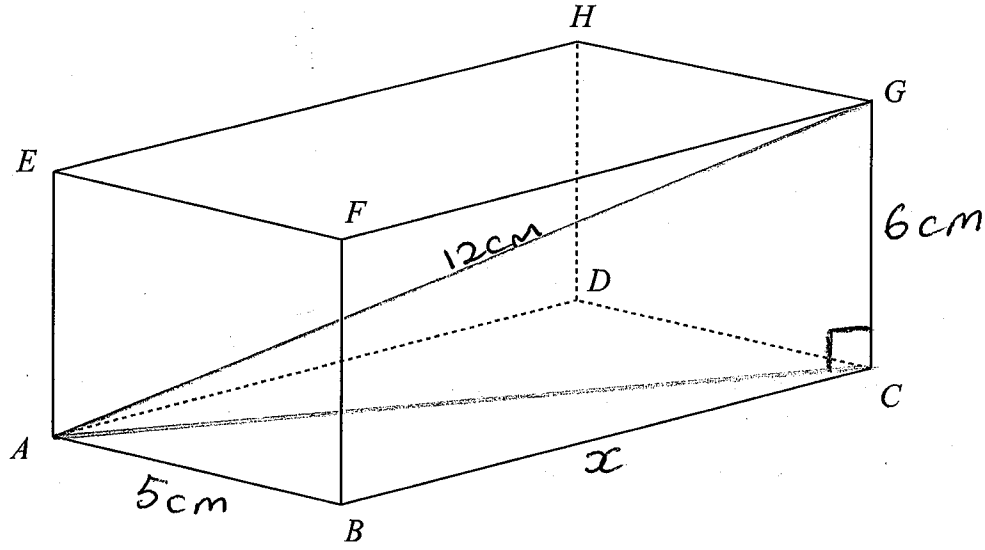
$$= \underline{10.2\text{ cm}}$$

10.2 cm

**(Total for Question 1 is 3 marks)**

**2** The diagram shows a cuboid *ABCDEFGH*.

$AB = 5$ cm
$AE = 6$ cm
$AG = 12$ cm




Calculate the length of *AD*.
Give your answer correct to 3 significant figures.

$$AC^2 + 6^2 = 12^2$$
$$AC^2 = 12^2 - 6^2$$
$$AC = \sqrt{12^2 - 6^2}$$
$$= 6\sqrt{3} \text{ cm}$$

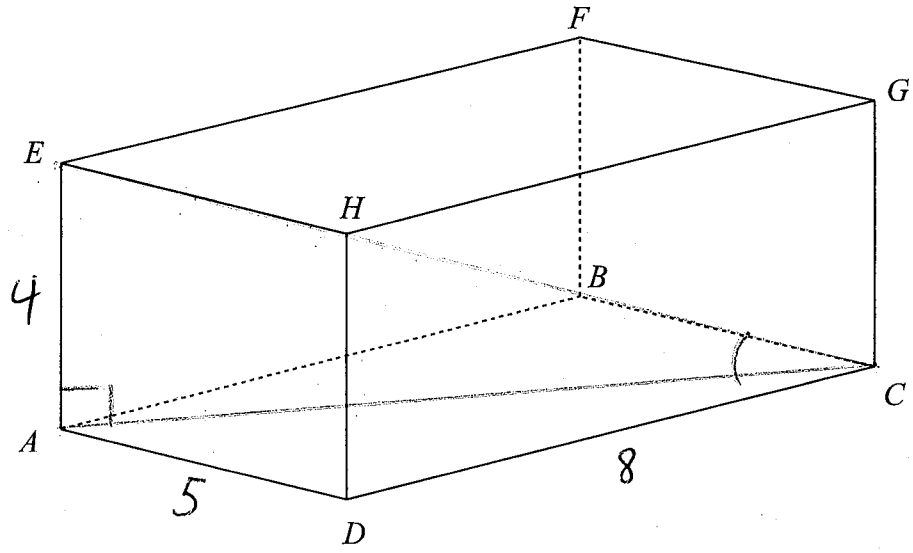
$$5^2 + x^2 = (6\sqrt{3})^2$$
$$x^2 = (6\sqrt{3})^2 - 5^2$$
$$= 108 - 25$$
$$x = \sqrt{83}$$
$$= 9.11$$

9.11 cm

**(Total for Question 2 is 4 marks)**

**3** The diagram shows a cuboid *ABCDEFGH*.

$AE = 4$ cm
$AD = 5$ cm
$DC = 8$ cm

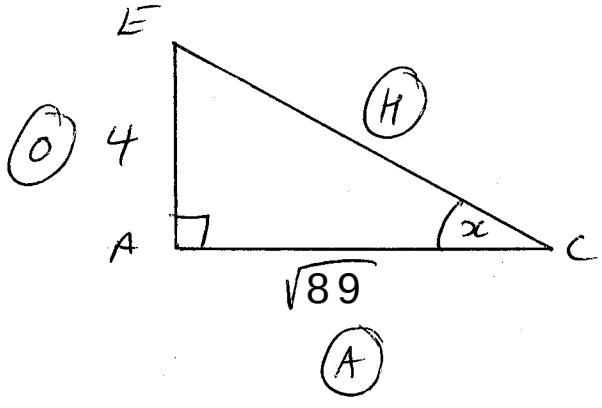



Calculate the size of angle *ECA*.
Give your answer correct to 3 significant figures.

$$AC^2 = 5^2 + 8^2$$

$$AC = \sqrt{5^2 + 8^2}$$

$$= \sqrt{89}$$




$$\tan x = \frac{O}{A}$$

$$= \frac{4}{\sqrt{89}}$$

$$x = \tan^{-1}\left(\frac{4}{\sqrt{89}}\right)$$

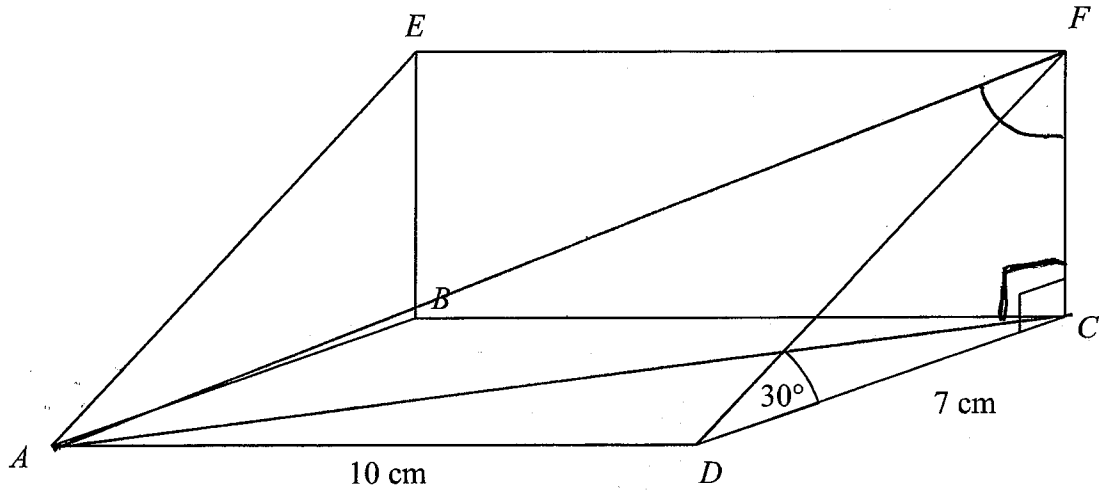
$$= 23.0^\circ$$

23.0 °

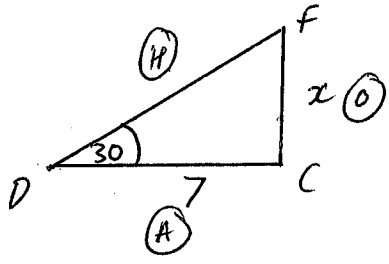
**(Total for Question 3 is 4 marks)**

**4** The diagram shows a triangular prism.

$CD = 7$ cm
$AD = 10$ cm
Angle $ADC = 30°$



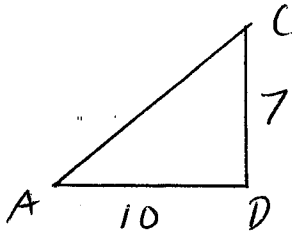

Calculate the size of angle $AFC$.
Give your answer correct to 1 decimal place.




$\tan 30 = \frac{x}{7}$

$x = 7 \tan 30$

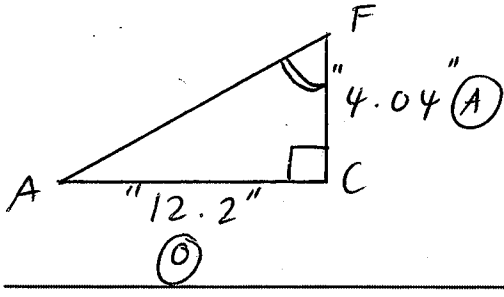
$= 4.04145...$




$AC^2 = 10^2 + 7^2$

$AC = \sqrt{10^2 + 7^2}$

$= 12.2065...$




$\tan \theta = \frac{12.2}{4.04}$

$\theta = \tan^{-1}\left(\frac{12.2}{4.04}\right)$

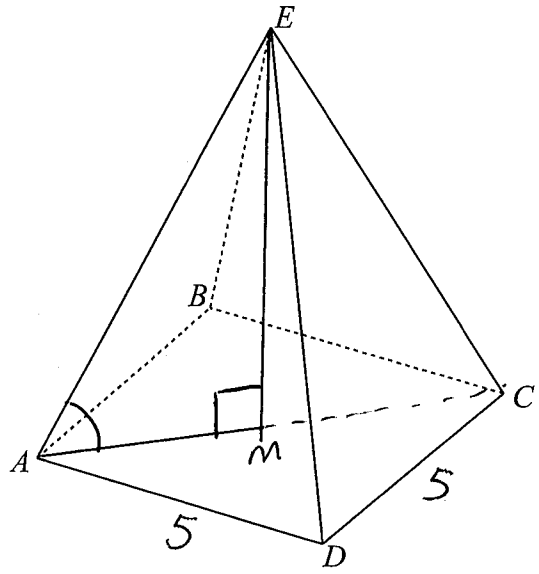
$= 71.7$

71.7 °

**(Total for Question 4 is 4 marks)**

**5** The diagram shows a pyramid.
The base of the pyramid *ABCD* is a square.

*AB* = 5 cm
The point *E* is 10 cm vertically above the base.




Calculate the size of angle *EAC*.

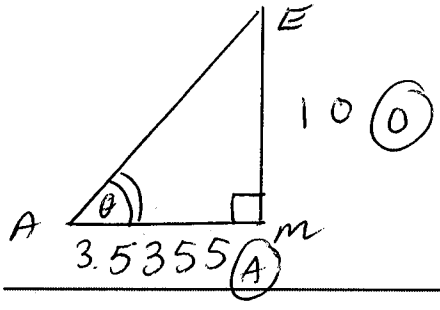
$AC^2 = 5^2 + 5^2$

$AC = \sqrt{5^2 + 5^2}$

$= 7.07106...$

$AM = \frac{7.07106}{2}$

$= 3.5355$




$\tan\theta = \frac{10}{3.5355}$

$\theta = \tan^{-1}\left(\frac{10}{3.5355}\right)$

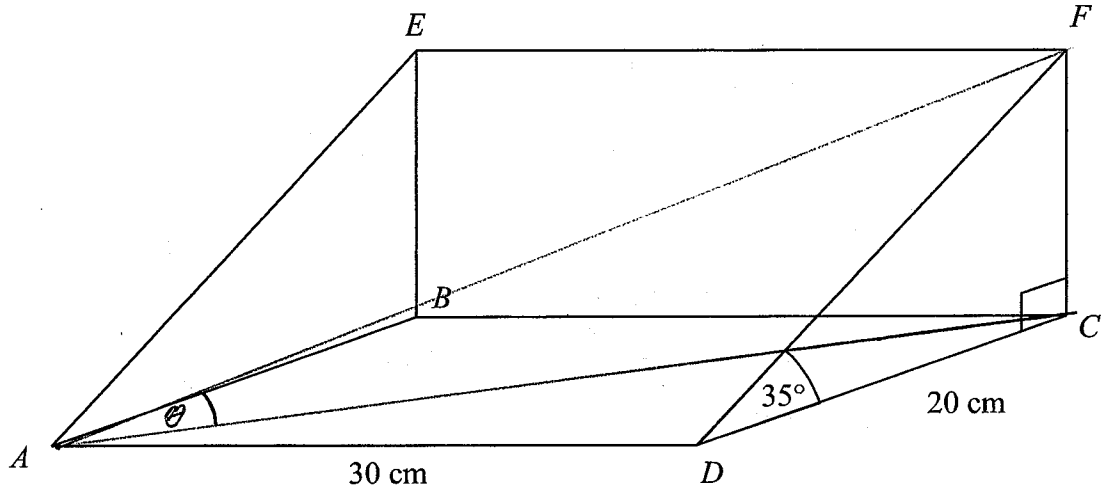
$= 70.5$

70.5 °

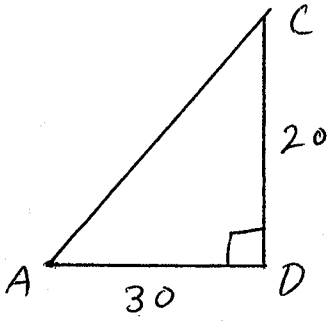
**(Total for Question 5 is 4 marks)**

**6** The diagram shows a triangular prism.

$CD = 20$ cm
$AD = 30$ cm
Angle $ADC = 35°$



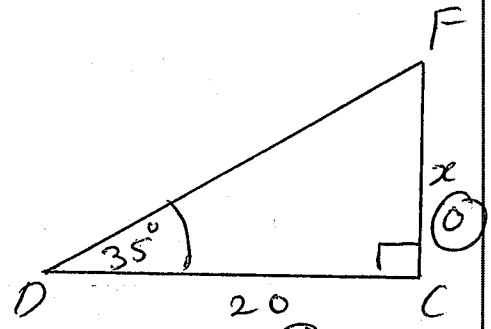

Calculate the size of angle the line $AF$ makes with the plane $ABCD$.
Give your answer correct to 3 significant figures.




$AC^2 = 30^2 + 20^2$

$AC = \sqrt{30^2 + 20^2}$

$= 36.0555$ cm

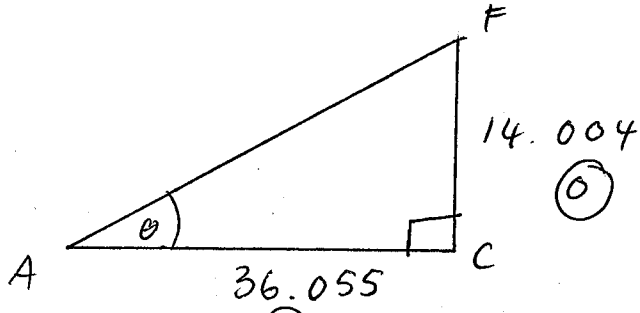



A

$\tan 35 = \frac{x}{20}$

$x = 20 \tan 35$

$= 14.004$




A

$\tan \theta = \frac{14.004}{36.055}$

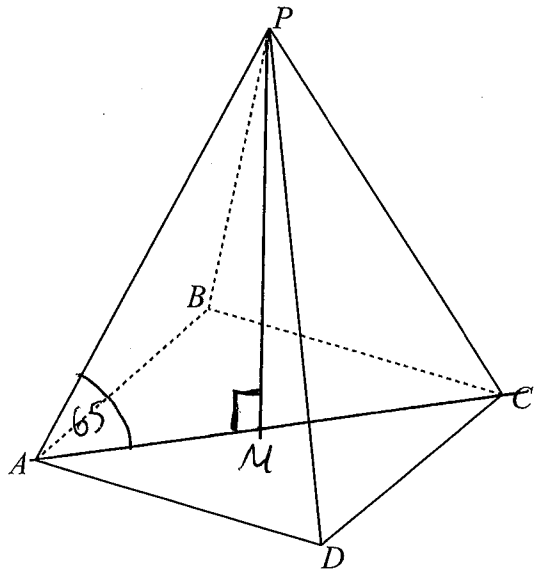
$= 21.2°$

21.2 °

**(Total for Question 6 is 4 marks)**

**7** The diagram shows a pyramid.
The base of the pyramid *ABCD* is a square.

*AB* = 15 cm
Angle *PAC* = 65°




Calculate the volume of the pyramid.

volume = $\frac{1}{3}$(base area) × height

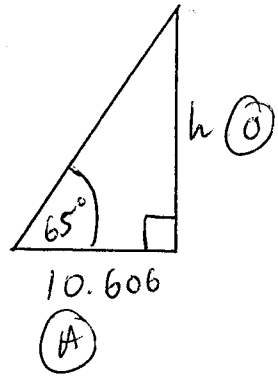
$AC^2 = 15^2 + 15^2$

$AC = \sqrt{15^2 + 15^2}$

$= 15\sqrt{2}$ cm

$AM = \frac{15\sqrt{2}}{2}$

$= 10.606$ cm




$\tan 65 = \frac{h}{10.606}$

$h = 10.6 \tan 65$

$= 22.74599\ldots$ cm

Area of base $= 15 \times 15$
$= 225$ cm$^2$

Volume $= \frac{1}{3}(225)(\text{"}22.7\text{"})$
$= 1706$

1706 cm$^3$

**(Total for Question 7 is 5 marks)**